

Chia-Heng Liu

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EDUCATION

Cornell University

Ithaca, NY

Master of Engineering in Electrical and Computer Engineering

Exp. Dec. 2026

Topic: Mixed-Precision Transformer Accelerator for LLM Inference

Advisor: Prof. Mohammad Alian

Coursework: Computer Architecture, Data Center Architecture, Computer Networks and Telecommunications

National Tsing Hua University (NTHU)

Hsinchu, Taiwan

Bachelor of Science in Materials Science and Engineering

Sep. 2020 – Jun. 2024

Coursework: Logic Design (A+), Computer Organization (A+), Electronics (A), Signals and Systems (A),

Semiconductor Design (A+)

SELECTED PROJECTS

ASIC Project: Stepwise-Streaming Systolic CNN Accelerator

System Verilog, Synopsys (VCS, Verdi, Design Compiler), ARM (SRAM Compiler), Pytorch *May. 2025 – Jul. 2025*

- Designed and implemented a CNN accelerator with a systolic array and cached weight dataflow, enabling cyclic weight reuse within PEs to minimize memory access.
- Applied tiling and scheduling with fused MAC–ReLU–Max operations, achieving **129 GFLOPs/s (INT8-equivalent) at 500 MHz**, and optimized parallel activation processing for higher efficiency.
- Integrated on-chip SRAM buffering and completed ASIC flow in Synopsys tools (Design Compiler, VCS, Verdi) targeting TSMC 40 nm, achieving **412.7 K μm^2** area; verified with RTL simulation and deployed a quantized MNIST model on a custom-ISA accelerator.

CPU Project: Fully-Bypassed Pipelined RISC-V CPU with Write-Back 2-Way Cache

SystemVerilog, Verilator, PyMTL3 *Sep. 2025*

- Designed and implemented a five-stage pipelined RISC-V CPU with hazard detection, stalling, and **full bypass forwarding logic**, achieving **near one-instruction-per-cycle throughput**.
- Integrated latency-insensitive val/rdy interfaces for multiplier and memory operations, and developed a **write-back 2-way set-associative cache** verified through PyMTL3 testbenches.

EXPERIENCE

Research Assistant at NYCU CERES Lab

Jul. 2024 – Aug. 2025

Publication (First Author): Energy-Efficient and Accurate Stochastic Computing-based

Multiply-Accumulate Architecture for Neural Network Accelerator Design

Feb. 2025 – Jul. 2025

IEEE International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoc), Dec. 2025

- Developed a stochastic-computing MAC architecture for energy-efficient DNN inference, synthesized in TSMC 40 nm (1049 μm^2 , 0.645 mW).
- Reduced MSE by 90.6% vs. prior SC MACs while maintaining 96.8% MNIST accuracy, matching binary baselines at lower cost.

Co-op Project with ITRI: NoC Accelerator Simulator Design and Low-Power Data Transmission Protocol, Movement, and Reuse Method

Jul. 2024 – Jan. 2025

- Built an AI accelerator simulator by integrating Gem5 (RISC-V CPU) and Noxim (NoC) with MCU-based scheduling and cache-interconnect modeling.
- Enhanced NoC with multi-layer, dependency-aware protocols to improve data reuse and identify bottlenecks on neural workloads.

TECHNICAL SKILLS

Hardware: Verilog, SystemVerilog, SystemC, RISC-V, Network-on-Chip, OpenPiton

EDA Tools: Synopsys (VCS, Verdi, Design Compiler, DesignWare), ARM (SRAM Compiler), Cadence (Innovus), Intel ModelSim, Xilinx Vivado, Verilator

Programming: C++, Python, Pytorch, RISC-V Assembly, Linux, Shell, TCL, LaTeX